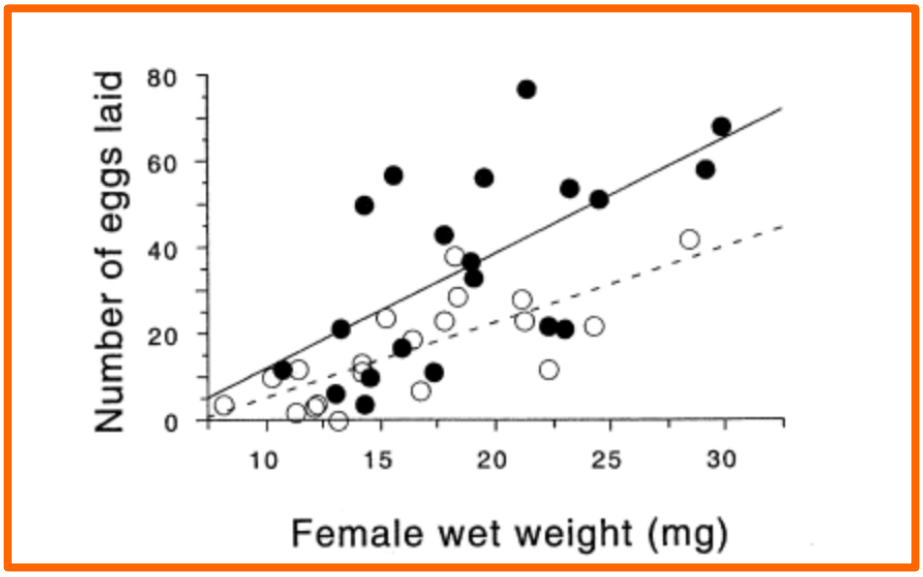
Module 4E Questions:

1. Females were mated once (white circles) or three times (black filled circles). Since larger females are known to produce more offspring, before mating, the female fireflies were weighed. The slopes between the once mated (dashed line) and thrice mated (dark line) were not significantly different but the Y-intercepts are significantly different. **Is there a difference in the number of eggs laid between once and thrice mated females? Why or why not?**



2. *We want to assess whether Diet affects gene expression, controlling for the influence of body weight.*

Source of variation	*Sum of Squares*	*df*	*Mean Square*	*F*	*P*
Body Weight	*50.2*	*1*	*50.2*	*12.8*	*0.001*
Diet	*20.6*	*2*	*10.3*	*2.63*	*0.091*
Residual	*78.0*	*19*	*4.11*		
Total	*148.8*	*22*			

   a. What is the covariate in this case?
   b. The first round of fitting has already occurred. **Write out the F equations for this second round and interpret the results of this table on gene expression.**

3. Researchers want to know whether **diet (Chow vs. HFD)** affects **Glucose12** after adjusting for **final weight (Weight12)**. They fit the model (Note: I am using the common notation of β_0 representing Y-intercept in this formula and Diet =HFD means that you are only investigating HFD results and not including chow diet):

$$\text{Glucose12} = \beta_0 + \beta_1(\text{Weight12}) + \beta_2(\text{Diet = HFD}) + \varepsilon$$

Below is a complete summary table of descriptive statistics so that you do **not** need to compute any sums of squares or means by hand.

Source	SS	df	MS	F	p-value
Weight12	**1117.3**	1	**1117.3**	255.7	< 0.0001
Diet	**13 357.5**	1	**13 357.5**	3055.3	< 0.0001
Residual	**39.3**	9	**4.37**	—	—
Total	**14 514.1**	11	—	—	—

Explain these results. Can you do two rounds of fitting? Why or why not.

4. Researchers want to know whether the relationship between **Weight12** and **Glucose12** differs across the three strains of mice (B6, BALB, CAST). They fit the following ANCOVA model:

$$\text{Glucose12} = \beta_0 + \beta_1(\text{Weight12}) + \beta_2(\text{Strain}) + \beta_3(\text{Weight12} \times \text{Strain}) + \varepsilon$$

   a. Using the ANCOVA table below, decide whether the **Weight12 × Strain interaction** should be kept or dropped.
   b. Explain your conclusion.

Source	SS	df	MS	F	p-value
Strain	291.4	2	145.7	33.41	0.0006
Weight12	**617.5**	1	617.5	141.6	<0.0001
Strain × Weight12	**7.1**	2	3.55	**0.81**	**0.49**
Residual (Error)	**26.2**	6	4.36	—	—
Total	942.2	11	—	—	—